Mammalia Raccoon Proximity Network Analysis

https://github.com/aiyshwary/Raccoon-Network-Analysis

Python **NetworkX** **Jupyter** **Graph Analysis** **Epidemiology**

OVERVIEW

A graph-based epidemiological study modelling rabies transmission risk in urban raccoon populations through proximity network analysis and centrality measures, supporting public health and wildlife management decisions.

IMPACT

Constructed and analyzed a raccoon proximity network using real-world contact data, applying centrality measures to identify high-risk individuals and quantify potential rabies spread pathways within urban populations.

TECHNICAL WALKTHROUGH

Mammalia Raccoon Proximity Network Analysis – Project

Explanation

Project Overview

This project analyzes social interaction patterns among urban raccoons using network analysis to understand how rabies and other pathogens can spread within raccoon populations.
Since raccoons are a major vector of rabies, studying how often and how closely they interact helps estimate disease transmission risk and supports better public health and wildlife management strategies.

Objective

Model raccoon interactions as a graph/network

Identify high-risk raccoons that can spread pathogens quickly Use centrality measures to understand social structure and disease flow

Dataset & Network Construction

24 raccoons → represented as nodes
~2000 interactions → represented as weighted edges

Data collected using proximity-detecting collars

Edge weight = time spent in close proximity Monthly interaction data used to build the network If two raccoons are connected, it means they had close physical interaction, which is critical for rabies transmission.
Why Network Analysis?
Disease spread is not random.

It depends on

i) Who interacts with whom

ii) How frequently

iii) Who acts as a bridge between groups

Network analysis helps identify super-spreaders and critical connectors.

Centrality Measures Used & Interpretation

Degree Centrality

i) Measures number of direct connections

ii) High degree = more interactions

Inference

i) Raccoon #4 has the highest degree

ii) Acts as a high-contact individual

iii) High risk of initiating widespread infection

Closeness Centrality

i) Measures how close a node is to all others

ii) Indicates how fast something can spread

Inference

i) Raccoon #2 has highest closeness centrality

ii) Can spread pathogens rapidly across the entire network

Betweenness Centrality

i) Measures how often a node lies on shortest paths

ii) Identifies bridge nodes

Inference

i) Raccoon #16 connects different subgroups

ii) Acts as a key transmission bridge

iii) Removing or vaccinating this raccoon can significantly slow disease spread

Clustering Coefficient

i) Measures how tightly a raccoon is embedded in a local group

ii) Indicates family or group behavior

Inference

i) Raccoon #2 has highest clustering

ii) Infection here may cause localized outbreaks within families

Eigenvector Centrality

i) Measures influence based on connections to influential nodes

Inference

i) Raccoon #4 again ranks highest

ii) Connected to other highly interactive raccoons

iii) Considered a super-spreader candidate

PageRank Algorithm

i) Evaluates overall importance considering direction and weight of interactions

Inference

i) Helps rank raccoons by global influence

Key Findings

i) Raccoon populations are highly interconnected

ii) Disease can spread much faster than expected

iii) A few raccoons play disproportionately large roles in pathogen transmission

iv) Targeted intervention (vaccination/tracking of key raccoons) is more effective than random control

Technologies & Concepts Used

i) Graph Theory

ii) Weighted graphs

iii) Centrality algorithms

iv) Epidemiological modeling concepts

KEY DETAILS

1. Built a weighted proximity network from the mammalia-raccoon-proximity dataset using Python and NetworkX.
2. Computed degree, betweenness, closeness, and eigenvector centrality to rank individuals by transmission risk.
3. Identified superspreader nodes whose removal would most disrupt disease propagation chains.
4. Visualized network topology and centrality distributions to communicate findings clearly.
5. Results provide evidence-based recommendations for targeted wildlife intervention strategies.

6. Analyzed a dense 24-node, 226-edge proximity graph (~82% density) derived from 1,997 weighted temporal contact records.

7. Computed clustering coefficients and aggregated per-node contact strength (total edge weight) to quantify interaction intensity — node 4 had the highest total contact weight (956,341).

8. Rendered the network using Kamada-Kawai force-directed layout (Matplotlib) to reveal spatial clustering patterns in the proximity data.